SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)*

QUESTIONS: 05 **Total Marks:100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I P.Vishwa Nandan /192473015*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

vishwanandan

Annexure A

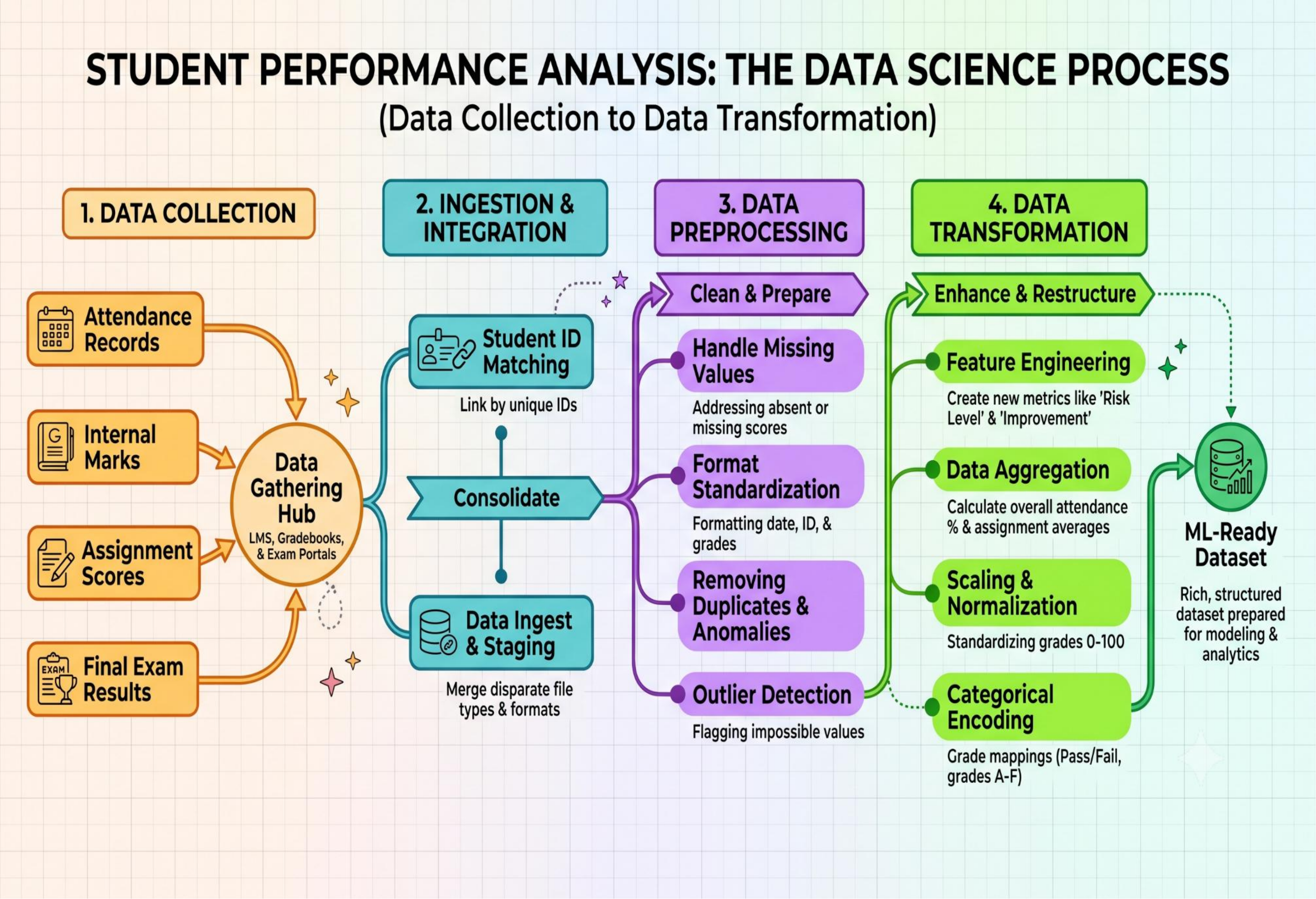
Concept Mapping

1. A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.
2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.
3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.
4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.
5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

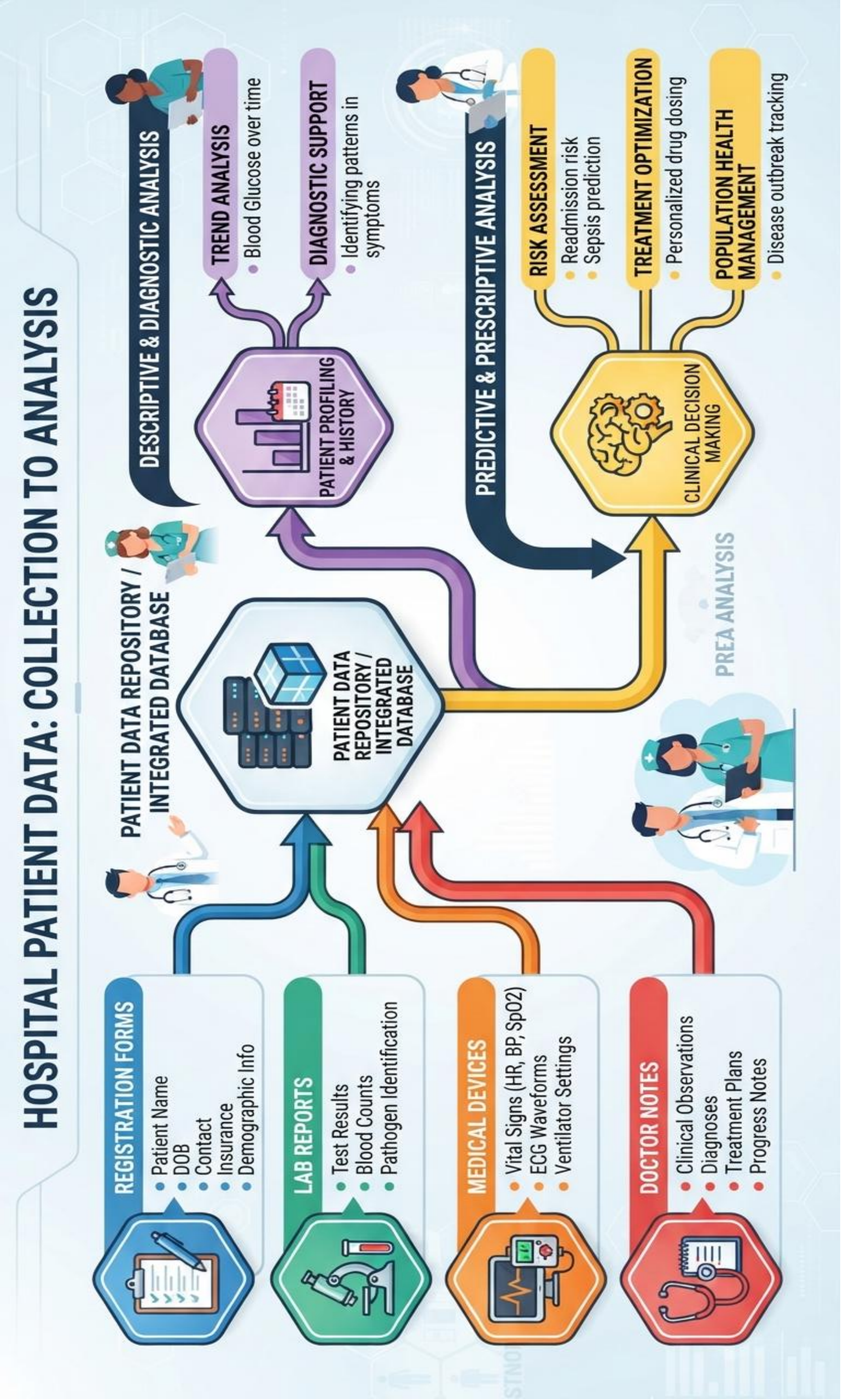
RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

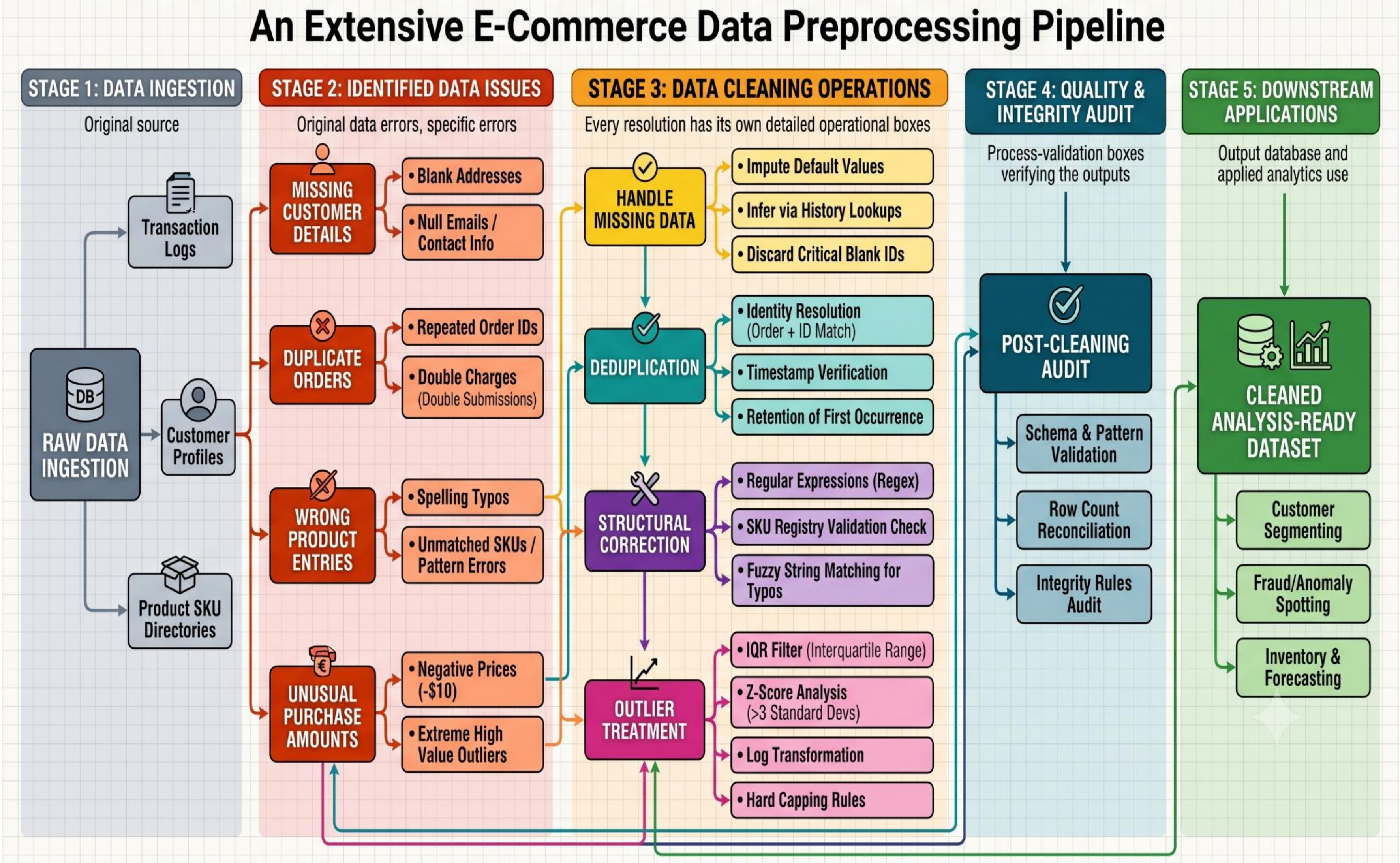
Concept Map 1


STUDENT PERFORMANCE ANALYSIS: THE DATA SCIENCE PROCESS
(Data Collection to Data Transformation)
1. DATA COLLECTION
Attendance Records
Internal Marks
Assignment Scores
Final Exam Results
EXAM
Data Gathering Hub
LMS, Gradebooks, & Exam Portals
2. INGESTION & INTEGRATION
Student ID Matching
Link by unique IDs
Consolidate
Data Ingest & Staging
Merge disparate file types & formats
3. DATA PREPROCESSING
Clean & Prepare
Handle Missing Values
Addressing absent or missing scores
Format Standardization
Formatting date, ID, & grades
Removing Duplicates & Anomalies
Outlier Detection
Flagging impossible values
4. DATA TRANSFORMATION
Enhance & Restructure
Feature Engineering
Create new metrics like 'Risk Level' & 'Improvement'
Data Aggregation
Calculate overall attendance % & assignment averages
Scaling & Normalization
Standardizing grades 0-100
Categorical Encoding
Grade mappings (Pass/Fail, grades A-F)
ML-Ready Dataset
Rich, structured dataset prepared for modeling & analytics

Concept Map 2


HOSPITAL PATIENT DATA: COLLECTION TO ANALYSIS
REGISTRATION FORMS
Patient Name
DOB
Contact
Insurance
Demographic Info
LAB REPORTS
Test Results
Blood Counts
Pathogen Identification
MEDICAL DEVICES
Vital Signs (HR, BP, SpO2)
ECG Waveforms
Ventilator Settings
DOCTOR NOTES
Clinical Observations
Diagnoses
Treatment Plans
Progress Notes
PATIENT DATA REPOSITORY / INTEGRATED DATABASE
PATIENT DATA REPOSITORY / INTEGRATED DATABASE
DESCRIPTIVE & DIAGNOSTIC ANALYSIS
PATIENT PROFILING & HISTORY
TREND ANALYSIS
Blood Glucose over time
DIAGNOSTIC SUPPORT
Identifying patterns in symptoms
PREDICTIVE & PRESCRIPTIVE ANALYSIS
CLINICAL DECISION MAKING
RISK ASSESSMENT
Readmission risk
Sepsis prediction
TREATMENT OPTIMIZATION
Personalized drug dosing
POPULATION HEALTH MANAGEMENT
Disease outbreak tracking
PREA ANALYSIS

Concept Map 3



An Extensive E-Commerce Data Preprocessing Pipeline

STAGE 1: DATA INGESTION

Original source

- RAW DATA INGESTION
  - Transaction Logs
  - Customer Profiles
  - Product SKU Directories

STAGE 2: IDENTIFIED DATA ISSUES

Original data errors, specific errors

- MISSING CUSTOMER DETAILS
  - Blank Addresses
  - Null Emails / Contact Info
- DUPLICATE ORDERS
  - Repeated Order IDs
  - Double Charges (Double Submissions)
- WRONG PRODUCT ENTRIES
  - Spelling Typos
  - Unmatched SKUs / Pattern Errors
- UNUSUAL PURCHASE AMOUNTS
  - Negative Prices (-$10)
  - Extreme High Value Outliers

STAGE 3: DATA CLEANING OPERATIONS

Every resolution has its own detailed operational boxes

- HANDLE MISSING DATA
  - Impute Default Values
  - Infer via History Lookups
  - Discard Critical Blank IDs
- DEDUPLICATION
  - Identity Resolution (Order + ID Match)
  - Timestamp Verification
  - Retention of First Occurrence
- STRUCTURAL CORRECTION
  - Regular Expressions (Regex)
  - SKU Registry Validation Check
  - Fuzzy String Matching for Typos
- OUTLIER TREATMENT
  - IQR Filter (Interquartile Range)
  - Z-Score Analysis (>3 Standard Devs)
  - Log Transformation
  - Hard Capping Rules

STAGE 4: QUALITY & INTEGRITY AUDIT

Process-validation boxes verifying the outputs

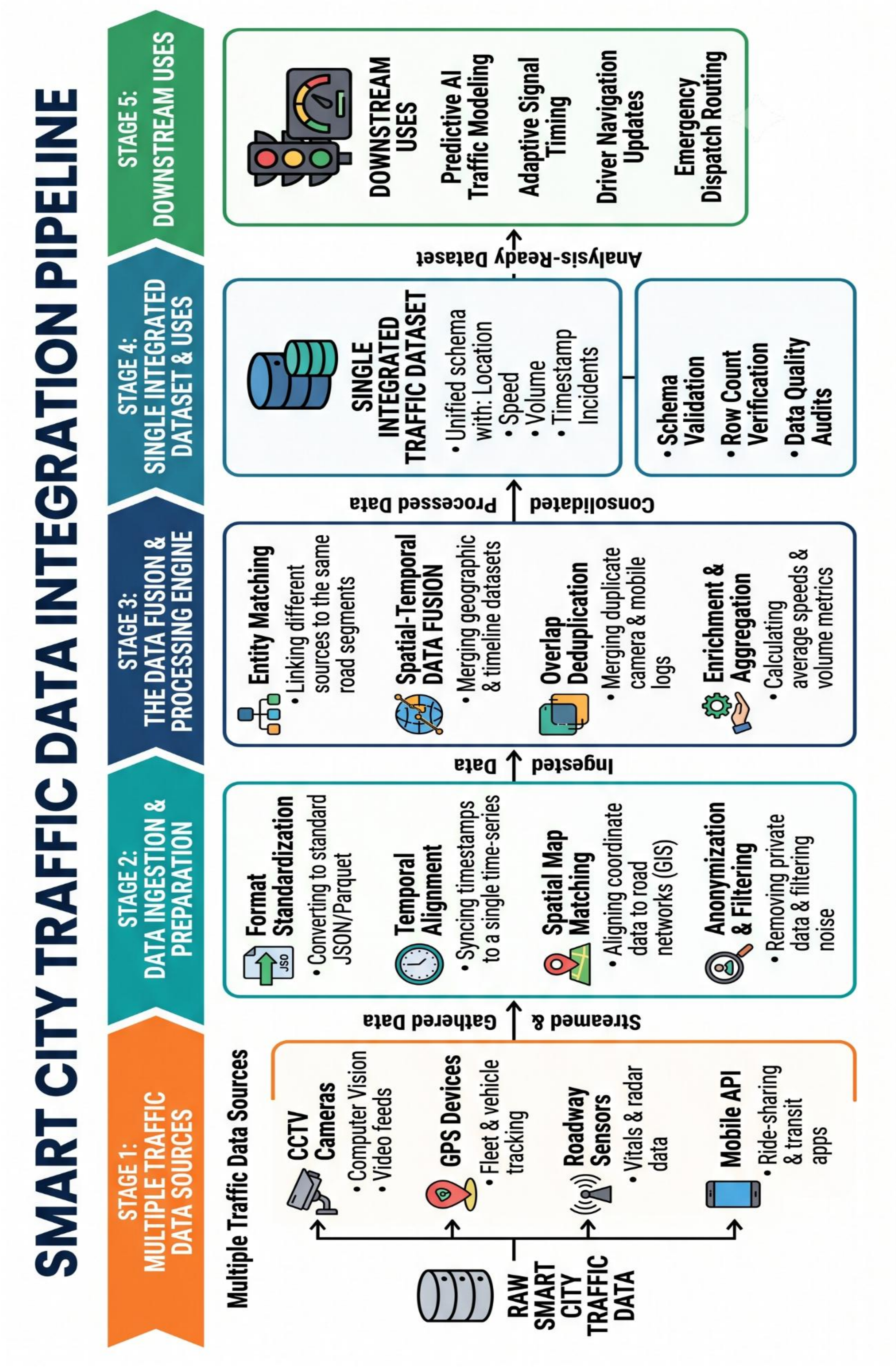
- POST-CLEANING AUDIT
  - Schema & Pattern Validation
  - Row Count Reconciliation
  - Integrity Rules Audit

STAGE 5: DOWNSTREAM APPLICATIONS

Output database and applied analytics use

- CLEANED ANALYSIS-READY DATASET
  - Customer Segmenting
  - Fraud/Anomaly Spotting
  - Inventory & Forecasting

Concept Map 4



SMART CITY TRAFFIC DATA INTEGRATION PIPELINE

STAGE 1: MULTIPLE TRAFFIC DATA SOURCES

RAW SMART CITY TRAFFIC DATA

Multiple Traffic Data Sources

CCTV Cameras
- Computer Vision
- Video feeds

GPS Devices
- Fleet & vehicle tracking

Roadway Sensors
- Vitals & radar data

Mobile API
- Ride-sharing & transit apps

Streamed & Gathered Data

STAGE 2: DATA INGESTION & PREPARATION

Format Standardization
- Converting to standard JSON/Parquet

Temporal Alignment
- Syncing timestamps to a single time-series

Spatial Map Matching
- Aligning coordinate data to road networks (GIS)

Anonymization & Filtering
- Removing private data & filtering noise

Ingested Data

STAGE 3: THE DATA FUSION & PROCESSING ENGINE

Entity Matching
- Linking different sources to the same road segments

Spatial-Temporal DATA FUSION
- Merging geographic & timeline datasets

Overlap Deduplication
- Merging duplicate camera & mobile logs

Enrichment & Aggregation
- Calculating average speeds & volume metrics

Consolidated Processed Data

STAGE 4: SINGLE INTEGRATED DATASET & USES

SINGLE INTEGRATED TRAFFIC DATASET
- Unified schema with: Location
  - Speed
  - Volume
  - Timestamp Incidents

- **Schema Validation**
- **Row Count Verification**
- **Data Quality Audits**

Analysis-Ready Dataset

STAGE 5: DOWNSTREAM USES

DOWNSTREAM USES

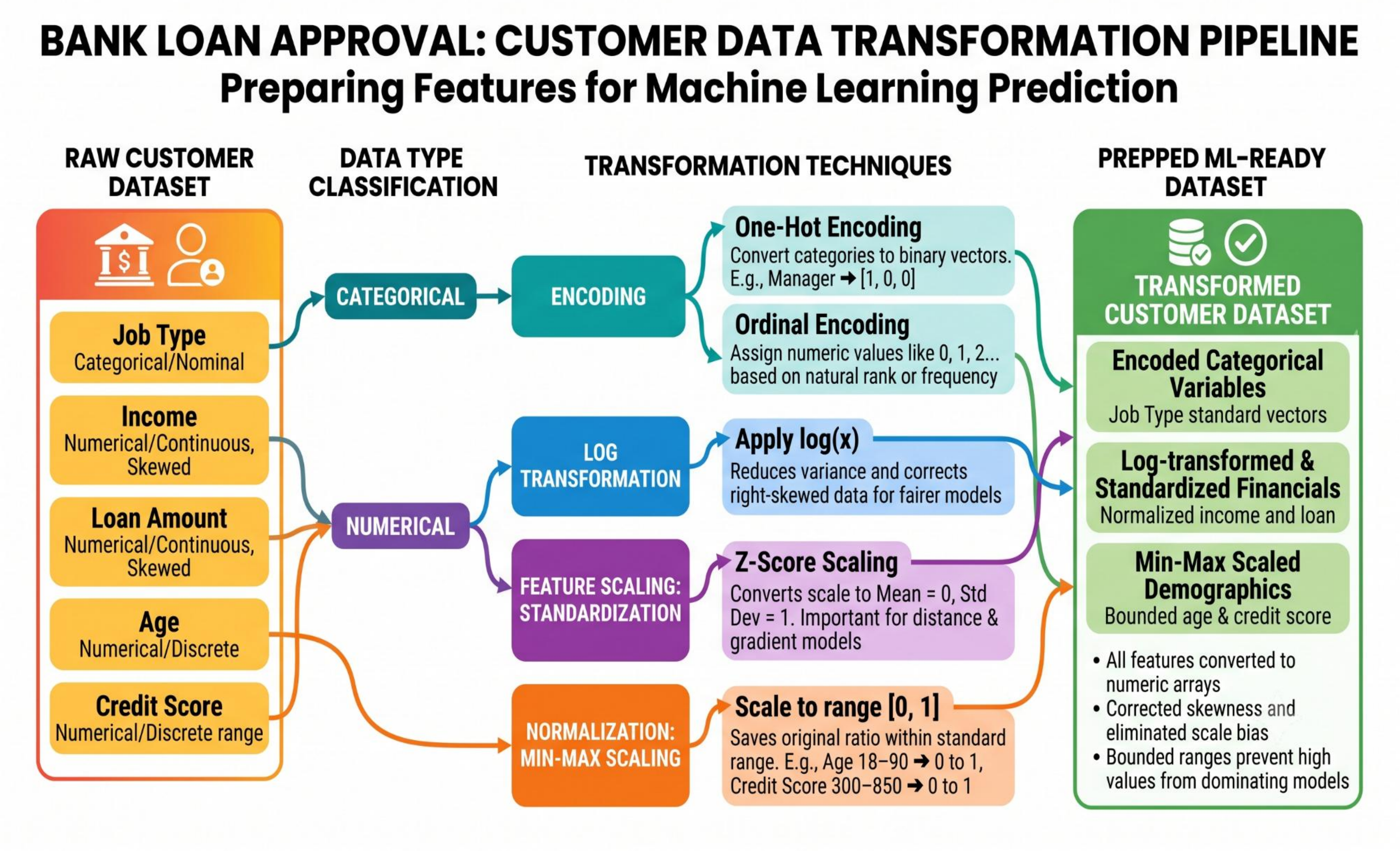
Predictive AI Traffic Modeling

Adaptive Signal Timing

Driver Navigation Updates

Emergency Dispatch Routing

Concept Map 5



BANK LOAN APPROVAL: CUSTOMER DATA TRANSFORMATION PIPELINE

Preparing Features for Machine Learning Prediction

RAW CUSTOMER DATASET

Job Type
Categorical/Nominal

Income
Numerical/Continuous, Skewed

Loan Amount
Numerical/Continuous, Skewed

Age
Numerical/Discrete

Credit Score
Numerical/Discrete range

DATA TYPE CLASSIFICATION

CATEGORICAL

NUMERICAL

TRANSFORMATION TECHNIQUES

ENCODING

LOG TRANSFORMATION

FEATURE SCALING: STANDARDIZATION

NORMALIZATION: MIN-MAX SCALING

One-Hot Encoding
Convert categories to binary vectors. E.g., Manager ➜ [1, 0, 0]

Ordinal Encoding
Assign numeric values like 0, 1, 2... based on natural rank or frequency

Apply log(x)
Reduces variance and corrects right-skewed data for fairer models

Z-Score Scaling
Converts scale to Mean = 0, Std Dev = 1. Important for distance & gradient models

Scale to range [0, 1]
Saves original ratio within standard range. E.g., Age 18–90 ➜ 0 to 1, Credit Score 300–850 ➜ 0 to 1

PREPPED ML-READY DATASET

TRANSFORMED CUSTOMER DATASET

Encoded Categorical Variables
Job Type standard vectors

Log-transformed & Standardized Financials
Normalized income and loan

Min-Max Scaled Demographics
Bounded age & credit score

- All features converted to numeric arrays
- Corrected skewness and eliminated scale bias
- Bounded ranges prevent high values from dominating models